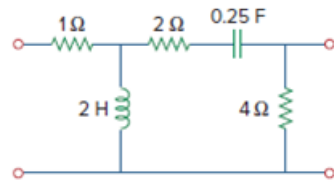


Problem

Use *PSpice* or *MultiSim* to determine the z parameters of the circuit in Fig. 19.125. Take $\omega = 2$ rad/s.



Step-by-step solution

Step 1 of 9

Draw the circuit diagram as shown below.

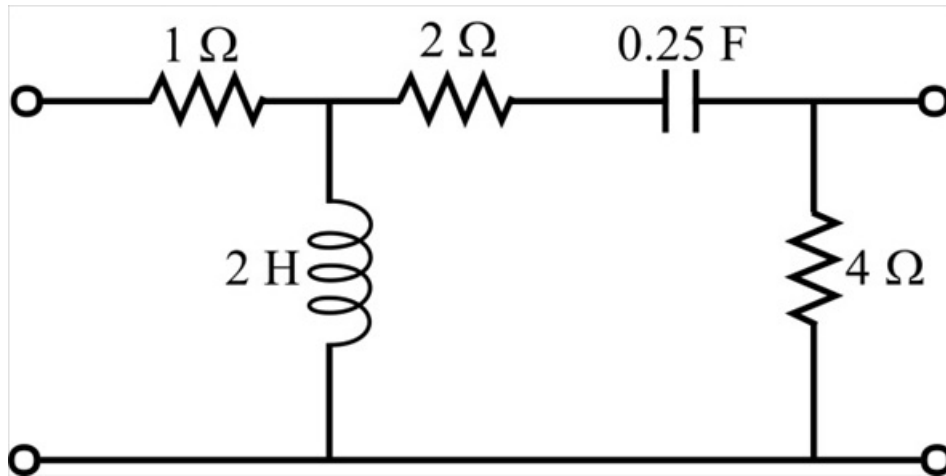


Figure 1

Step 2 of 9

Re-draw Figure 1, with node parameters.

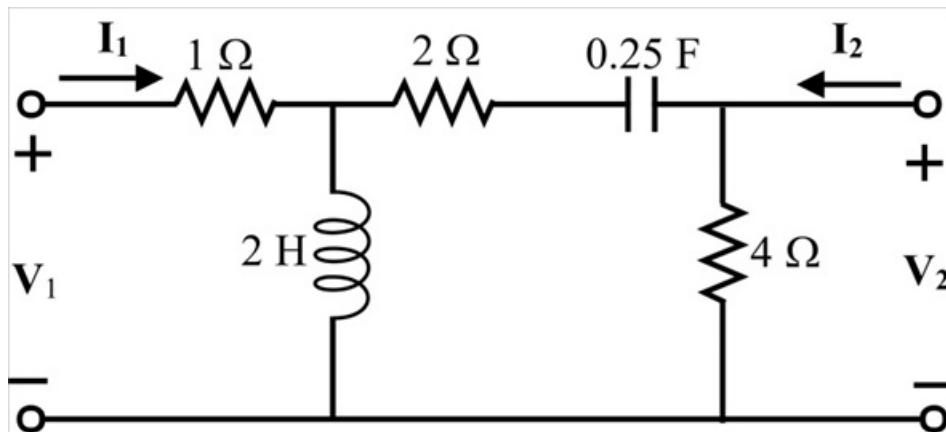


Figure 2

Step 3 of 9

Consider, $\omega = 2 \text{ rad/s}$.

$$\begin{aligned} f &= \frac{\omega}{2\pi} \\ &= \frac{2}{2\pi} \\ &= 0.3183 \text{ Hz} \end{aligned}$$

Calculate z -parameters.

$$[z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} \quad Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} \quad Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

Step 4 of 9

Calculate Z_{11} and Z_{21} .

(i) Set $I_1 = 1 \text{ A}$, $I_2 = 0$, start freq = 0.3183 Hz and End freq = 0.3183 Hz

Draw Pspice to obtain Z_{11} and Z_{21} .

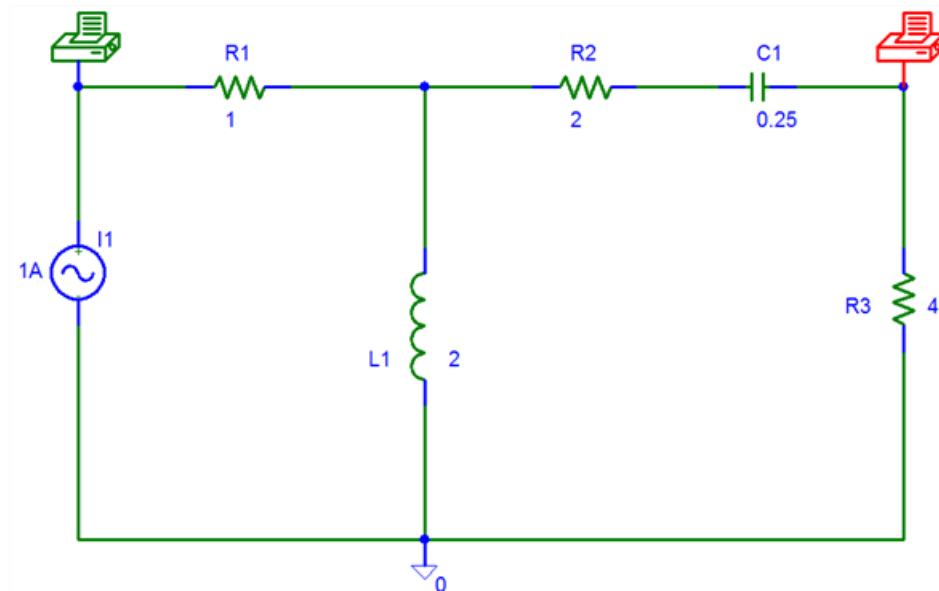


Figure 3

Step 5 of 9

From Figure 3, the following Pspice results are obtained.

FREQ VM(\$N_0001) VP(\$N_0001)

3.183E-01 4.669E+00 -1.367E+02

FREQ VM(\$N_0004) VP(\$N_0004)

3.183E-01 2.530E+00 -1.084E+02

From the results of Pspice, Calculate z_{11} and z_{21} values.

Calculate z_{11} value.

$$\begin{aligned} z_{11} &= \frac{V_1}{I_1} \\ &= \frac{4.669 \angle -136.7^\circ}{1} \\ &= 4.669 \angle -136.7^\circ \Omega \end{aligned}$$

Therefore, the value of z_{11} is $4.669 \angle -136.7^\circ \Omega$.

Calculate z_{21} value.

$$\begin{aligned} z_{21} &= \frac{V_2}{I_1} \\ &= \frac{2.53 \angle -108.4^\circ}{1} \\ &= 2.53 \angle -108.4^\circ \Omega \end{aligned}$$

Therefore, the value of z_{21} is $2.53 \angle -108.4^\circ \Omega$.

Step 6 of 9

Calculate z_{12} and z_{22} .

(ii) Set $I_2 = 1$ A, $I_1 = 0$. start freq = 0.1592 Hz and End freq = 0.1592 Hz.

Draw Pspice to obtain z_{12} and z_{22} .

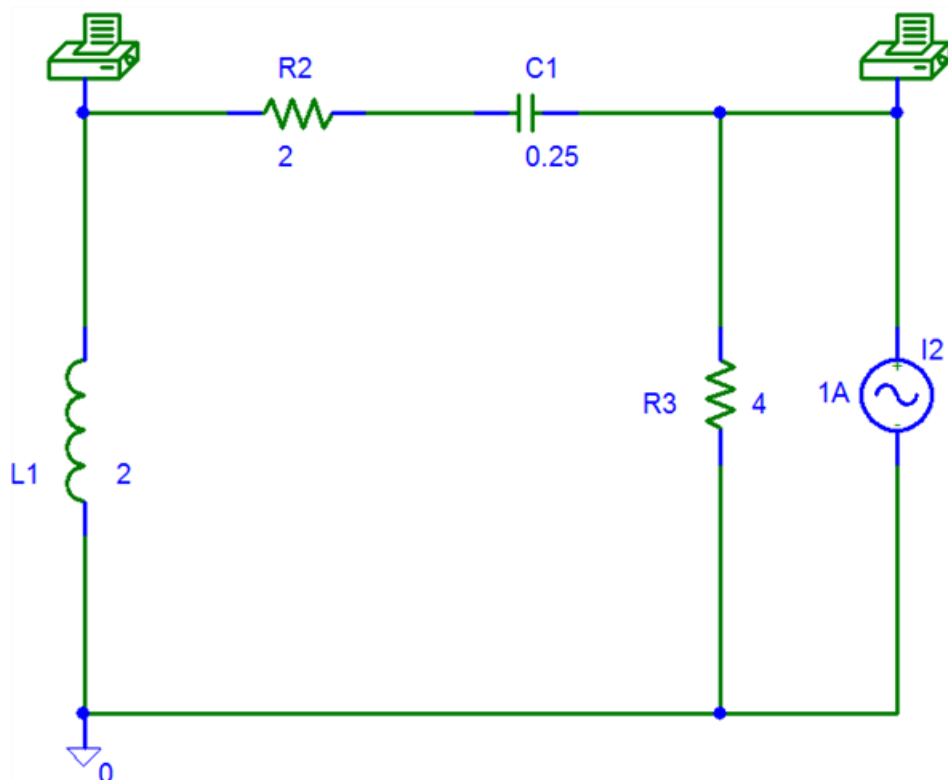


Figure 4

Step 7 of 9

From Figure 4, the following Pspice results are obtained.

FREQ VM(\$N_0003) VP(\$N_0003)

3.183E-01 1.789E+00 -1.534E+02

FREQ VM(\$N_0002) VP(\$N_0002)

3.183E-01 2.530E+00 -1.084E+02

Step 8 of 9

From the results of Pspice, Calculate z_{12} and z_{22} values.

Calculate z_{12} value.

$$\begin{aligned} z_{12} &= \frac{V_1}{I_2} \\ &= \frac{2.53 \angle -108.4^\circ}{1} \\ &= 2.53 \angle -108.4^\circ \Omega \end{aligned}$$

Therefore, the value of z_{12} is $2.53 \angle -108.4^\circ \Omega$.

Calculate z_{22} value.

$$\begin{aligned} z_{22} &= \frac{V_2}{I_2} \\ &= \frac{1.789 \angle -153.4^\circ}{1} \\ &= 1.789 \angle -153.4^\circ \Omega \end{aligned}$$

Therefore, the value of z_{22} is $1.789 \angle -153.4^\circ \Omega$.

Step 9 of 9

Therefore, the z -parameters are:

$$[z] = \begin{bmatrix} 4.669 \angle -136.7^\circ & 2.53 \angle -108.4^\circ \\ 2.53 \angle -108.4^\circ & 1.789 \angle -153.4^\circ \end{bmatrix} \Omega$$